

Sparse-Reference 5G CIR Localization via Self-Supervised Representation Learning and Channel Charting

1. Overview and Task Understanding

IPIN 2026 Competition Track 7 focuses on two-dimensional indoor localization using 5G channel impulse responses (CIRs) with sparse position references. The official dataset contains CIR measurements collected by eight distributed base stations from 5G uplink sounding reference signals. For each sample, the competing system is expected to estimate the two-dimensional position of the user equipment (UE).

The main difficulty of this track is the sparse-reference setting. Only 50 samples are provided with valid ground-truth coordinates, while most entries in the reference field are unavailable. Therefore, the task cannot be treated as a conventional fully supervised regression problem. A successful system should exploit the large amount of unlabeled CIR data and learn a robust radio-domain representation that can be aligned to the physical coordinate system using only a small number of labeled references.

We propose an offline localization pipeline based on self-supervised CIR representation learning and channel charting. The system first learns latent radio features from all available CIR samples, including unlabeled data. It then constructs a low-dimensional radio chart that preserves local neighborhood relationships in the wireless channel space. Finally, the sparse reference samples are used to align the learned chart or latent representation to the physical two-dimensional coordinate system.

The complete system consists of six main stages: data preprocessing, CIR feature construction, self-supervised representation learning, channel charting, sparse reference alignment, and trajectory-level robust post-processing. Since Track 7 is an offline task, the system will process the competition dataset in batch mode and output two-dimensional position estimates according to the official submission format.

2. Data Preprocessing and Feature Construction

The official dataset is provided in HDF5 format and contains a dataset named `ds`. Each sample includes CIR, toa, antenna_mask, timestamp, and reference. The CIR field has shape (2, 8, 128), where the first dimension represents the real and imaginary parts of the complex CIR, the second dimension corresponds to the eight base stations, and the third dimension corresponds to 128 delay taps.

For each sample, we combine the real and imaginary components into a complex-valued CIR matrix with shape 8×128 . From the complex CIR, we construct log-amplitude profiles, normalized energy profiles, peak-delay features, peak-energy features, and delay-spread statistics. Optional phase-related features may also be evaluated. Since the CIR amplitude is not normalized and may be affected by UE transmit power adaptation, different normalization strategies will be compared,

including per-sample normalization, per-base-station normalization, and energy-preserving normalization.

The `antenna_mask` field is explicitly used throughout the pipeline. Invalid base-station observations are not treated as valid zero-valued measurements. Instead, the mask is propagated to feature aggregation, neural encoding, and loss computation. This mask-aware design enables the model to handle missing or degraded base-station measurements.

The `toa` values are used as auxiliary features rather than direct ranging measurements. According to the technical annex, CIR windows may sometimes be incorrectly selected in severe multipath or signal degradation conditions, which can also produce incorrect ToA values. Therefore, the proposed system does not rely on pure ToA-based trilateration. Abnormal ToA patterns will be detected and down-weighted when constructing auxiliary features.

The `timestamp` field is used to identify continuous recording segments. Since the dataset is composed of multiple recordings, timestamp discontinuities are used to split the data into separate trajectory segments. Temporal operations, including contrastive pairing, trajectory smoothing, and speed-constrained filtering, are only applied within the same continuous recording.

3. Self-Supervised CIR Representation Learning

Because only 50 labeled reference samples are available, directly training a high-capacity supervised localization model may lead to severe overfitting. To address this issue, we use all labeled and unlabeled samples to pretrain a CIR encoder with self-supervised objectives. The purpose of this stage is to learn a latent representation that captures stable multipath structures, inter-base-station relationships, and local trajectory continuity.

The CIR encoder processes the delay-tap sequence of each base station using a lightweight one-dimensional convolutional network or Transformer-style architecture. The per-base-station representations are then aggregated through mask-aware pooling or attention, so that invalid base-station measurements do not contribute incorrectly to the global representation. The encoder outputs a compact latent feature vector for each CIR sample.

The first self-supervised objective is masked CIR reconstruction. Random delay taps and, in some cases, entire base-station branches are masked during training. An encoder-decoder model is trained to reconstruct the missing log-amplitude CIR profiles. This objective encourages the encoder to learn meaningful radio-channel structures and correlations across delay taps and base stations, rather than memorizing the limited labeled samples.

The second self-supervised objective is temporal contrastive learning. Within each continuous recording, temporally close samples are treated as positive pairs because the UE is mounted on a mobile robot and nearby timestamps usually correspond to nearby physical positions. Samples far apart in time or from different recording segments are treated as negative pairs. A contrastive loss encourages

neighboring radio observations to have neighboring embeddings, which supports the subsequent channel charting stage.

In parallel, non-neural baseline methods will also be implemented using engineered CIR statistics, PCA or UMAP dimensionality reduction, and K-nearest-neighbor regression. These baselines will be used to validate the preprocessing pipeline and provide robust fallback estimators.

4. Channel Charting and Sparse Reference Alignment

After self-supervised pretraining, the learned CIR representations are used to construct a low-dimensional channel chart. The goal of channel charting is to preserve local neighborhood relationships in the radio-domain representation without requiring dense position labels. Ideally, samples that are close in physical space should also be close in the learned radio chart.

We will evaluate several charting strategies, including UMAP-based charting, PCA-based baselines, autoencoder-based charting, and direct use of the pretrained encoder latent space. The chart may be constructed as a two-dimensional representation or as a slightly higher-dimensional intermediate representation followed by coordinate mapping.

The sparse reference samples are then used to align the radio chart to the physical coordinate system. Since only 50 labeled coordinates are available, we prioritize low-complexity and robust alignment methods. Candidate methods include K-nearest-neighbor regression, radial basis function interpolation, thin-plate spline alignment, Procrustes-based alignment, and lightweight multilayer perceptron regression heads.

To reduce overfitting, hyperparameters will be selected using leave-one-out validation or repeated sparse-label validation on the 50 reference samples. Model complexity will be controlled through regularization, early stopping, and simple non-parametric mappings. The final prediction may combine several complementary estimators, such as a chart-based interpolator and a regularized regression head trained on top of the pretrained CIR encoder.

5. Robust Processing for Multipath, Signal Loss, and Outlier

Observations

The recording environment is a railway maintenance facility with many metallic structures, including train compartments, access platforms, and maintenance equipment. Such an environment causes strong reflection, scattering, non-line-of-sight propagation, and signal distortion. Therefore, the system treats the CIR as a multipath fingerprint rather than relying on a single direct path or a pure ToA-based range estimate.

Missing or invalid base-station observations are handled using `antenna_mask`. Invalid measurements are excluded from mask-aware feature aggregation and loss computation. For samples with severe signal degradation, the system may assign

lower confidence to the corresponding radio features and rely more on neighboring temporal constraints during post-processing.

ToA outliers and incorrectly selected CIR windows are handled through consistency checks. Abnormal ToA values, inconsistent energy profiles, and sudden changes in peak-delay features will be detected and down-weighted. ToA is used as supportive information rather than as a hard geometric constraint.

Trajectory-level post-processing is used to reduce physically implausible position jumps. Since the UE is mounted on a robot with a maximum speed of approximately 1.5 m/s, adjacent predictions within the same continuous recording should satisfy a reasonable motion constraint:

$$\|\hat{P}_{t+1}-\hat{P}_t\|_2 \leq 1.5\Delta t + \epsilon$$

where $\hat{P}_t$ denotes the estimated two-dimensional position at time t , Δt is the time difference between adjacent samples, and ϵ is a tolerance term. We will evaluate moving-average smoothing, constant-velocity Kalman filtering, and graph-based trajectory smoothing. All temporal smoothing is restricted to samples within the same recording segment and is not applied across timestamp discontinuities.

6. Evaluation Criteria and Implementation Plan

The official point-wise error metric is the Euclidean distance between the estimated and true two-dimensional positions:

$$e_i = \|\hat{P}_i - P_i\|_2$$

where $\hat{P}_i$ is the estimated position and P_i is the ground-truth position. The final ranking metric is the third quartile of the error distribution, i.e., the 75th percentile positioning error. Therefore, the 75th percentile Euclidean error will be used as the primary model-selection criterion.

During internal validation, we will monitor mean error, median error, 75th percentile error, and outlier rate. Since only 50 labeled samples are available, we will use leave-one-out validation and repeated sparse-label splits to compare candidate models and select hyperparameters. Candidate systems include engineered-feature baselines, self-supervised encoders with regression heads, channel charting with non-parametric alignment, and ensemble combinations.

The system will be implemented in Python using standard scientific computing and machine learning libraries, including h5py, NumPy, SciPy, scikit-learn, UMAP, and PyTorch. As Track 7 is an offline task, the final system will process the competition dataset in batch mode and generate two-dimensional position estimates for all required samples according to the official submission format.